

BIOGEOCHEMICAL CYCLES AND ECOLOGICAL PYRAMIDS

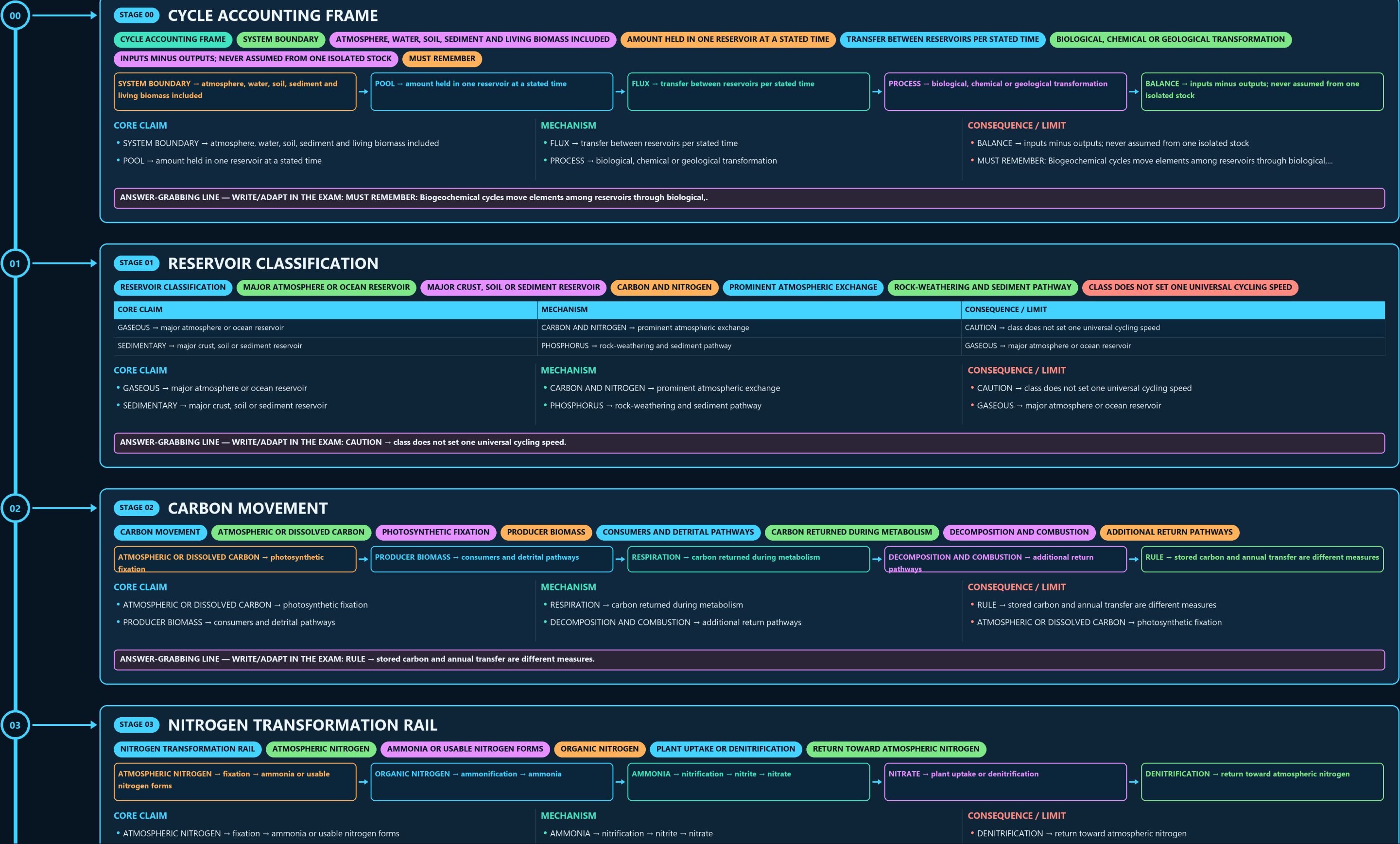
CYCLE ACCOUNTING FRAME → RESERVOIR CLASSIFICATION → CARBON MOVEMENT → NITROGEN TRANSFORMATION RAIL → NITROGEN ORGANISM MAP → PHOSPHORUS PATH

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

• ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

• PRODUCER BIOMASS → consumers and detrital pathways

• RESPIRATION → carbon returned during metabolism

• DECOMPOSITION AND COMBUSTION → additional return pathways

• RULE → stored carbon and annual transfer are different measures

• ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → stored carbon and annual transfer are different measures.

03

STAGE 03

NITROGEN TRANSFORMATION RAIL

NITROGEN TRANSFORMATION RAIL

ATMOSPHERIC NITROGEN

AMMONIA OR USABLE NITROGEN FORMS

ORGANIC NITROGEN

PLANT UPTAKE OR DENITRIFICATION

RETURN TOWARD ATMOSPHERIC NITROGEN

ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ORGANIC NITROGEN → ammonification → ammonia

AMMONIA → nitrification → nitrite → nitrate

NITRATE → plant uptake or denitrification

DENITRIFICATION → return toward atmospheric nitrogen

CORE CLAIM

• ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

• ORGANIC NITROGEN → ammonification → ammonia

MECHANISM

• AMMONIA → nitrification → nitrite → nitrate

• NITRATE → plant uptake or denitrification

CONSEQUENCE / LIMIT

• DENITRIFICATION → return toward atmospheric nitrogen

• ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms.

04

STAGE 04

NITROGEN ORGANISM MAP

NITROGEN ORGANISM MAP

SYMBIOTIC FIXATION ASSOCIATED WITH LEGUME ROOTS

FREE-LIVING FIXATION EXAMPLE IN THE OWNER

AMMONIA TO NITRITE STEP

NITRITE TO NITRATE STEP

DENITRIFICATION ROUTE

AZOTOBACTER → free-living fixation example in the owner

RHIZOBIUM → symbiotic fixation associated with legume roots

NITROSOMONAS → ammonia to nitrite step

NITROBACTER → nitrite to nitrate step

CORE CLAIM

• RHIZOBIUM → symbiotic fixation associated with legume roots

• AZOTOBACTER → free-living fixation example in the owner

MECHANISM

• NITROSOMONAS → ammonia to nitrite step

• NITROBACTER → nitrite to nitrate step

CONSEQUENCE / LIMIT

• PSEUDOMONAS-TYPE → denitrification route

• RHIZOBIUM → symbiotic fixation associated with legume roots

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AZOTOBACTER → free-living fixation example in the owner.

05

STAGE 05

PHOSPHORUS PATH

PHOSPHORUS PATH

PHOSPHATE ROCK

SOIL OR WATER PHOSPHATE

PLANT UPTAKE

CONSUMER TRANSFER

DETRITAL RETURN

AQUATIC LOADING

MINING AND FERTILISER

PHOSPHATE ROCK → weathering → soil or water phosphate

PLANT UPTAKE → consumer transfer → detrital return

RUNOFF → aquatic loading → sedimentation

MINING AND FERTILISER → accelerate extraction and dispersal

BOUNDARY → no significant atmospheric phase in the owner

CORE CLAIM

• PHOSPHATE ROCK → weathering → soil or water phosphate

• PLANT UPTAKE → consumer transfer → detrital return

MECHANISM

• RUNOFF → aquatic loading → sedimentation

• MINING AND FERTILISER → accelerate extraction and dispersal

CONSEQUENCE / LIMIT

• BOUNDARY → no significant atmospheric phase in the owner

• PHOSPHATE ROCK → weathering → soil or water phosphate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → no significant atmospheric phase in the owner.

06

STAGE 06

CYCLE-SPECIFIC HUMAN PRESSURE

CYCLE-SPECIFIC HUMAN PRESSURE

FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE

SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN

MINING, FERTILISER APPLICATION AND RUNOFF

CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER

SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

CORE CLAIM

CARBON → fossil-fuel combustion and land-cover change

NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

MECHANISM

PHOSPHORUS → mining, fertiliser application and runoff

OUTCOMES → climate forcing, nutrient pollution and resource loss differ

CONSEQUENCE / LIMIT

POLICY → source control must match the cycle and reservoir

CARBON → fossil-fuel combustion and land-cover change

CORE CLAIM

• CARBON → fossil-fuel combustion and land-cover change

• NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

MECHANISM

• PHOSPHORUS → mining, fertiliser application and runoff

• OUTCOMES → climate forcing, nutrient pollution and resource loss differ

CONSEQUENCE / LIMIT

• POLICY → source control must match the cycle and reservoir

• CARBON → fossil-fuel combustion and land-cover change

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06

STAGE 06

CYCLE-SPECIFIC HUMAN PRESSURE

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FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE

SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN

MINING, FERTILISER APPLICATION AND RUNOFF

CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER

SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CARBON → fossil-fuel combustion and land-cover change	PHOSPHORUS → mining, fertiliser application and runoff	POLICY → source control must match the cycle and reservoir
NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen	OUTCOMES → climate forcing, nutrient pollution and resource loss differ	CARBON → fossil-fuel combustion and land-cover change

CORE CLAIM

- CARBON → fossil-fuel combustion and land-cover change
- NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

MECHANISM

- PHOSPHORUS → mining, fertiliser application and runoff
- OUTCOMES → climate forcing, nutrient pollution and resource loss differ

CONSEQUENCE / LIMIT

- POLICY → source control must match the cycle and reservoir
- CARBON → fossil-fuel combustion and land-cover change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → source control must match the cycle and reservoir.

07

STAGE 07

EUTROPHICATION MECHANISM

EUTROPHICATION MECHANISM

NUTRIENT ENRICHMENT

INCREASED ALGAL OR PLANT PRODUCTION

ORGANIC MATERIAL ACCUMULATES

DECOMPOSER ACTIVITY RISES

OXYGEN DEMAND RISES

DISSOLVED OXYGEN MAY DECLINE

ECOLOGICAL STRESS

NUTRIENT ENRICHMENT → increased algal or plant production

→

ORGANIC MATERIAL ACCUMULATES → decomposer activity rises

→

OXYGEN DEMAND RISES → dissolved oxygen may decline

→

ECOLOGICAL STRESS → composition changes and mortality may occur

→

CLASSIFY → natural ageing versus human-accelerated cultural eutrophication

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
- NUTRIENT ENRICHMENT → increased algal or plant production - ORGANIC MATERIAL ACCUMULATES → decomposer activity rises	- OXYGEN DEMAND RISES → dissolved oxygen may decline - ECOLOGICAL STRESS → composition changes and mortality may occur	- CLASSIFY → natural ageing versus human-accelerated cultural eutrophication - NUTRIENT ENRICHMENT → increased algal or plant production

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLASSIFY → natural ageing versus human-accelerated cultural eutrophication.

08

STAGE 08

THREE PYRAMID PARAMETERS

THREE PYRAMID PARAMETERS

COUNT OF ORGANISMS AT EACH TROPHIC LEVEL

STANDING LIVING MATERIAL AT A STATED TIME

TRANSFER THROUGH TROPHIC LEVELS OVER TIME

DEPENDS ON PARAMETER AND ECOSYSTEM

ONLY THE ENERGY PYRAMID IS ALWAYS UPRIGHT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NUMBERS → count of organisms at each trophic level	ENERGY → transfer through trophic levels over time	RULE → only the energy pyramid is always upright
BIOMASS → standing living material at a stated time	SHAPE → depends on parameter and ecosystem	NUMBERS → count of organisms at each trophic level

CORE CLAIM

- NUMBERS → count of organisms at each trophic level
- BIOMASS → standing living material at a stated time

MECHANISM

- ENERGY → transfer through trophic levels over time
- SHAPE → depends on parameter and ecosystem

CONSEQUENCE / LIMIT

- RULE → only the energy pyramid is always upright
- NUMBERS → count of organisms at each trophic level

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → only the energy pyramid is always upright.

09

STAGE 09

AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION

SMALL PHYTOPLANKTON STANDING CROP

RAPID REPLACEMENT

HIGH TURNOVER

CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD

CONSUMER STANDING BIOMASS

MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT

ENERGY TRANSFER

SMALL PHYTOPLANKTON STANDING CROP → rapid replacement

→

HIGH TURNOVER → continued production over the measurement period

→

CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant

→

ENERGY TRANSFER → still declines upward

→

VERDICT → biomass inversion is not energy inversion

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement - HIGH TURNOVER → continued production over the measurement period	- CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant - ENERGY TRANSFER → still declines upward	- VERDICT → biomass inversion is not energy inversion - CLOSE DISTINCTION: Reservoir stock is not flux, residence time is not turnover rate,...

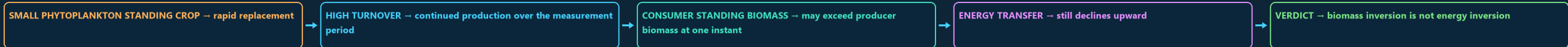
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

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09

STAGE 09 AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION SMALL PHYTOPLANKTON STANDING CROP RAPID REPLACEMENT HIGH TURNOVER CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD CONSUMER STANDING BIOMASS MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT ENERGY TRANSFER



CORE CLAIM

- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement
- HIGH TURNOVER → continued production over the measurement period

MECHANISM

- CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant
- ENERGY TRANSFER → still declines upward

CONSEQUENCE / LIMIT

- VERDICT → biomass inversion is not energy inversion
- CLOSE DISTINCTION: Reservoir stock is not flux, residence time is not turnover rate,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

10

STAGE 10 HEURISTIC AND MODEL FIREWALL

HEURISTIC AND MODEL FIREWALL ENERGY LOSS QUALITATIVE THERMODYNAMIC DIRECTION IS ROBUST TEN-PER-CENT RULE OWNER LABELS IT AN AVERAGE HEURISTIC WEAKENS A RIGID ONE-LEVEL ASSIGNMENT CHANGES STANDING STOCKS AND FEEDING RELATIONS

USEFUL SUMMARY, NOT A COMPLETE FOOD WEB

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ENERGY LOSS → qualitative thermodynamic direction is robust	OMNIVORY → weakens a rigid one-level assignment	PYRAMID → useful summary, not a complete food web
TEN-PER-CENT RULE → owner labels it an average heuristic	SEASONALITY → changes standing stocks and feeding relations	EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name pool, process, direction, limiting...

CORE CLAIM

- ENERGY LOSS → qualitative thermodynamic direction is robust
- TEN-PER-CENT RULE → owner labels it an average heuristic

MECHANISM

- OMNIVORY → weakens a rigid one-level assignment
- SEASONALITY → changes standing stocks and feeding relations

CONSEQUENCE / LIMIT

- PYRAMID → useful summary, not a complete food web
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name pool, process, direction, limiting...

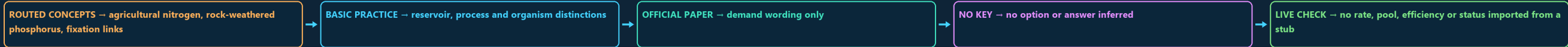
MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name pool, process, direction, limiting...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEN-PER-CENT RULE → owner labels it an average heuristic.

11

STAGE 11 PYQ AND EVIDENCE BOUNDARY

PYQ AND EVIDENCE BOUNDARY ROUTED CONCEPTS AGRICULTURAL NITROGEN, ROCK-WEATHERED PHOSPHORUS, FIXATION LINKS BASIC PRACTICE RESERVOIR, PROCESS AND ORGANISM DISTINCTIONS OFFICIAL PAPER DEMAND WORDING ONLY NO KEY



CORE CLAIM

- ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links
- BASIC PRACTICE → reservoir, process and organism distinctions

MECHANISM

- OFFICIAL PAPER → demand wording only
- NO KEY → no option or answer inferred

CONSEQUENCE / LIMIT

- LIVE CHECK → no rate, pool, efficiency or status imported from a stub
- ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIVE CHECK → no rate, pool, efficiency or status imported from a stub.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT CPCB MONITORS WATER-QUALITY PARAMETERS (BOD, DISSOLVED OXYGEN, NUTRIENT LOAD) MINISTRY OF AGRICULTURE AND FARMERS WELFARE DEPARTMENT OF FERTILIZERS IMPLEMENTATION GAP FERTILISER SUBSIDY DESIGN IN INDIA HAS HISTORICALLY FAVOURED CARBON-CYCLE SCIENCE (IPCC AR6, 2023) IS THE SCIENTIFIC SPINE NATIONAL-SCALE NUTRIENT-CYCLE DISRUPTION (E.G., TOTAL AGRICULTURAL NITROGEN RUNOFF) IS

OPTIONAL SOURCE DEPTH

- CPCB monitors water-quality parameters (BOD, dissolved oxygen, nutrient load) that
- Ministry of Agriculture and Farmers Welfare / Department of Fertilizers: regulate
- CAUTION: Implementation gap: fertiliser subsidy design in India has historically favoured
- CAUTION: Carbon-cycle science (IPCC AR6, 2023) is the scientific spine of the entire UNFCCC/Paris

ADVANCED DEBATE

- CAUTION: National-scale nutrient-cycle disruption (e.g., total agricultural nitrogen runoff) is
- CAUTION: "Ecological efficiency of 10%" is an average heuristic; actual transfer efficiency
- Gaseous cycles (carbon, nitrogen, oxygen, water) self-regulate faster than sedimentary
- Biological nitrogen fixation converts atmospheric N₂ to ammonia; industrial fixation via

LEGACY / NUANCE

- The pyramid of energy is always upright due to progressive energy loss at each trophic
- Eutrophication depletes dissolved oxygen as decomposer microbial activity spikes to
- NOT: Nutrient pollution and carbon emissions are governed by the same policy lever. - They
- NOT: An inverted biomass pyramid means an unhealthy ecosystem. - It reflects fast nutrient

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.